Dr Lewis A. Jones

NERC Independent Research Fellow
Department of Earth Sciences
University College London, London, UK

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Research summary

Palaeobiologist examining the macroecological and macroevolutionary history of reef-building organisms, and marine invertebrate biodiversity more broadly, through the integration of interdisciplinary tools and diverse sources of information, such as ecological modelling, Earth System modelling, and fossil and extant occurrence datasets. An additional significant theme of research has been evaluating the influence of data incompleteness on perceptions of the geological past, such as biodiversity trends, and the development of software tools for the research community. Current research focus is investigating the drivers of marine biodiversity hotspots throughout the Cenozoic and reconstructing their evolutionary history.

Macroecology and biodiversity | **Fossil record bias** | **Corals and reef ecosystems** | **Palaeoclimate** | **R programming**

Education

PhD in Palaeobiology (2016–2020)
Department of Earth Science and Engineering, Imperial College London, London, UK
Research advisors: Prof. Philip D. Mannion & Prof. Peter A. Allison
Thesis title: On the detectability of latitudinal biodiversity gradients in deep time

MPhysGeog in Physical Geography (2011–2015)
Department of Geography and Earth Science, University of Hull, Hull, UK
Research advisor: Prof. David P.G. Bond
Thesis title: The End-Triassic mass extinction. An investigation of the British Geological Record: testing for the role of volcanism and ocean anoxia

Chartered Management Institute (Level 5) in Management and Leadership (2012–2015)
Leeds University Officer Training Corps, Leeds, UK

Employment

NERC Independent Research Fellow (2024–current)
Department of Earth Sciences, University College London, London, UK

Juan de la Cierva Postdoctoral Research Fellow (2023–2024)
Mapas Lab, Centro de Investigación Mariña, Universidade de Vigo, Vigo, Spain

European Research Council Postdoctoral Research Assistant (2021–2022)
Mapas Lab, Centro de Investigación Mariña, Universidade de Vigo, Vigo, Spain

Graduate Teaching Assistant (2016–2021)
Department of Earth Science and Engineering, Imperial College London, London, UK

Student Hall's Pastoral Warden (2016–2021)
Woodward Buildings, Imperial College London, London, UK

British Army Officer Cadet (2012–2015)
Leeds University Officer Training Corps, Leeds, UK

Sales Assistant (2010–2012)
Republic Ltd, Grimsby and Hull, UK

Baker (2008–2010)
Muffin Break, Grimsby, UK

Research visits

University of Vienna, Vienna, Austria (April–June, 2023)
Visiting researcher in the Conservation Palaeobiology and Historical Ecology Lab (PI: Prof. Martin Zuschin)

Publications

Peer-reviewed publications

1. **Lewis A. Jones**, Christopher D. Dean, Bethany J. Allen, Harriet B. Drage, Joseph T. Flannery-Sutherland, William Gearty, Alfio A. Chiarenza, Erin M. Dillon, Bruna M. Farina, and Pedro L. Godoy (2025). Ten simple rules to follow when cleaning occurrence data in palaeobiology. *Palaeontology* 68(5), e70028. DOI: **10.1111/pala.70028**.
2. Christopher D. Dean, Alfio A. Chiarenza, Jeffrey W. Doser, Alexander Farnsworth, **Lewis A. Jones**, Sinéad J. Lyster, Charlotte L. Outhwaite, Paul J. Valdes, Richard J. Butler, and Philip D. Mannion (2025). The structure of the end-Cretaceous dinosaur fossil record in North America. *Current Biology* 35 (9), 1973–1988 . DOI: **10.1016/j.cub.2025.03.025**.
3. Theres Koch, Angelina Ivkić, **Lewis A. Jones**, Victor S. Scharnhorst, Constanze M. Stix, Antonia Auer, Theda Schöchtner, Gözde Özer, Simon Steinwender, Joseph W. Daurella, and Martin Zuschin (2025). Quantifying diversity and growth form distribution of scleractinian corals, in Mangrove Bay, El Qoseir, Egypt. *Biodiversity Data Journal* 13, e134282. DOI: **10.3897/BDJ.13.e134282**.
4. **Lewis A. Jones**, Christopher D. Dean, William Gearty, and Bethany J. Allen (2024). rmacrostrat: An R package for accessing and retrieving data from the Macrostrat geological database. *Geosphere* 20(6), 1456–1467. DOI: **10.1130/GES02815.1**.
5. **Lewis A. Jones** and Mathew Domeier (2024). A Phanerozoic gridded dataset for palaeogeographic reconstructions. *Scientific Data* 11 (710), 1–7. DOI: **10.1038/s41597-024-03468-w**.
6. Alfio A. Chiarenza, Juan L. Cantalapiedra, **Lewis A. Jones**, Sara Gamboa, Sofía Galván, Alexander Farnsworth, Paul J. Valdes, Graciela Sotelo, and Sara Varela (2024). Early Jurassic origin of avian endothermy and thermophysiological diversity in dinosaurs. *Current Biology* 34 (11), 2518–2527. DOI: **10.1016/j.cub.2024.04.051**.
7. Kilian Eichenseer and **Lewis A. Jones** (2024). Bayesian multi-proxy reconstruction of early Eocene latitudinal temperature gradients. *Climate of the Past* 20, 349–362. DOI: **10.5194/cp-20-349-2024**.
8. Lucas Buffan, **Lewis A. Jones**, Mathew Domeier, Christopher R. Scotese, Sabin Zahirovic and Sara Varela (2023). Mind the uncertainty: Global Plate Model choice impacts deep-time palaeobiological studies. *Methods in Ecology and Evolution* 14 (12), 3007–3019. DOI:**10.1111/2041-210X.14204**.
9. William Gearty and **Lewis A. Jones** (2023). rphylopic: An R package for fetching, transforming, and visualising PhyloPic silhouettes. *Methods in Ecology and Evolution* 14 (11), 2700–2708. DOI:**10.1111/2041-210X.14221**.
10. **Lewis A. Jones**, William Gearty, Bethany J. Allen, Kilian Eichenseer, Christopher D. Dean, Sofía Galván, Miranta Kouvari, Pedro L. Godoy, Cecily Nicholl, Lucas Buffan, Erin M. Dillon, Joseph T. Flannery-Sutherland and A. Alessandro Chiarenza (2023). palaeoverse: A community-driven R package to support palaeobiological analysis. *Methods in Ecology and Evolution* 14 (9), 2205–2215. DOI:**10.1111/2041-210X.14099**.
11. **Lewis A. Jones**, Philip D. Mannion, Alexander Farnsworth, Fran Bragg, and Daniel J. Lunt (2022). Climatic and tectonic drivers shaped the tropical distribution of coral reefs. *Nature Communications* 13 (3120), 1–10. DOI:**10.1038/s41467-022-30793-8**.
12. **Lewis A. Jones** and Kilian Eichenseer (2022). Uneven spatial sampling distorts reconstructions of Phanerozoic seawater temperature. *Geology* 50 (2), 238–242. DOI:**10.1130/G49132.1**.
13. **Lewis A. Jones**, Christopher D. Dean, Philip D. Mannion, Alexander Farnsworth, and Peter A. Allison (2021). Spatial sampling heterogeneity limits the detectability of deep time latitudinal biodiversity gradients. *Proceedings of the Royal Society B* 288 (1945), 20202762. DOI:**10.1098/rspb.2020.2762**.
14. **Lewis A. Jones**, Philip D. Mannion, Alexander Farnsworth, Paul Valdes, Sarah-Jane Kelland, and Peter A. Allison (2019). Coupling of palaeontological and neontological reef coral data improves forecasts of biodiversity responses under global climatic change. *Royal Society Open Science* 6 (4), 182111. DOI:**10.1098/rsos.182111**.
15. Alfio Alessandro Chiarenza, Philip D. Mannion, Daniel Lunt, Alexander Farnsworth, **Lewis A. Jones**, Sarah-Jane Kelland, and Peter A. Allison (2019). Ecological niche modelling does not support climatically-driven dinosaur diversity decline before the

Cretaceous/Paleogene mass extinction. *Nature Communications* 10 (1091), 1–14. DOI:**10.1038/s41467-019-08997-2**.

In-press and submitted manuscripts

1. **Lewis A. Jones**, Danijela Dimitrijević, Markus Aretz, Francesca R. Bosellini, Juan C. Braga, Clara Diaz, Elizabeth Dowding, Kenneth Johnson, Wolfgang Kiessling, Bernard Lathuilière, Anna Merkel, Kyle Morgan, Michele Morsilli, John M. Pandolfi, Mikołaj K. Zapalski, and Nadia Santodomingo (In review). The potential of mesophotic environments to provide refugia for coral communities during times of crises. *Nature Communications*.
2. **Lewis A. Jones**, William Gearty, Lucas Buffan, and Bethany J. Allen (In review). Global plate model choice impacts reconstructions of the latitudinal biodiversity gradient. *Global Change Ecology*. Preprint DOI: **10.1101/2025.01.09.632144**.
3. Angelina Ivkić, **Lewis A. Jones**, Andreas Kroh, Abbas Mansour, Mohamed Osman, Mohamed Hassan, and Martin Zuschin (In review). Reconstructing ecosystem baselines using Pleistocene data: insights from an Egyptian coral reef. *Coral Reefs*.
4. Miranta Kouvari, Kateryn Pino, **Lewis A. Jones**, Juan Carrillo, Grace Varnham, Alexander Farnsworth, Anjali Goswami, and Philip Mannion (In review). The impact of Andean uplift and climate on eutherian mammal diversification and the emergence of the present-day latitudinal biodiversity gradient in South America. *Current Biology*.

Theses

1. **Lewis A. Jones** (2020). On the detectability of latitudinal biodiversity gradients in deep time. PhD Thesis. Imperial College London. Supervisors: Prof. Philip D. Mannion and Prof. Peter A. Allison. DOI:**10.25560/100239**.
2. **Lewis A. Jones** (2015). The end-Triassic mass extinction. An investigation of the British geological record: testing for the role of volcanism and ocean anoxia. Master Thesis. Supervisor: Prof. David Bond.

Other works

1. **Lewis A. Jones**, Christopher D. Dean, and Harriet B. Drage (2025). **Introducing the Palaeoverse**. *The Palaeontological Association Newsletter* 118.
2. **Lewis A. Jones** (2024). **Meeting report: R for Palaeobiologists: Workshop and Hackathon**. *The Palaeontological Association Newsletter* 116.
3. **Lewis A. Jones** (2023). **Palaeontology's greatest ever graphs: The absence of evidence: spatial sampling bias in the fossil record**. *The Palaeontological Association Newsletter* 114.
4. **Lewis A. Jones** (2022). **Book review: Timefulness: How thinking like a geologist can help save the world**. *The Palaeontological Association Newsletter* 110.
5. Michael Stevens, Yiqiao Chen, Alex Stringer, Caitlin Clemmow, and **Lewis A. Jones** (2020). **Key factors driving obesity in the UK**. *GISRUK* 2020.
6. **Lewis A. Jones** (2020). **Meeting report: 13th International Symposium of Fossil Cnidaria and Porifera**. *The Palaeontological Association Newsletter* 102.

Published software

1. **palaeoverse**: A community-driven R package to support data preparation and exploration for palaeobiological analyses. CRAN: **https://cran.r-project.org/web/packages/palaeoverse/**. GitHub: **https://github.com/palaeoverse/palaeoverse**.
2. **rphylopic**: An R package for fetching, transforming, and visualising PhyloPic silhouettes. CRAN: **https://cran.r-project.org/web/packages/rphylopic/**. GitHub: **https://github.com/palaeoverse/rphylopic**.
3. **sepkoski**: A data R package of Sepkoski's Fossil Marine Animal Genera Compendium (Sepkoski, 2002). CRAN: **https://cran.r-project.org/web/packages/sepkoski/**. GitHub: **https://github.com/lewisajones/sepkoski**.
4. **rmacrostrat:** An R package to fetch geologic data from the Macrostrat database. CRAN: **https://cran.r-project.org/web/packages/rmacrostrat/**. GitHub: **https://github.com/palaeoverse/rmacrostrat**.

Academic awards and grants

Geared for Success: Enhancing Fieldwork Equity in Earth Science Education (2026)
Take Bold Action for Inclusion, Faculty of Mathematical & Physical Sciences, University College London, UK
Role: applicant, value: 2400 GBP (+ 2400 GBP match funding from the Department of Earth Sciences)

Converting Users to Contributors: Enabling Sustainable Maintenance and Development of Palaeoverse (2025)
Research Software Maintenance Fund, Software Sustainability Institute, UK
Role: applicant, value: 634,129 GBP

Open Science and Scholarship Awards, University College London
Runner-up award for 'Enhancing open science and reproducibility capacity in the academic community'
Benthic marine ecosystem dynamics through time and space (BENEDICTS) (2025)
Proyectos de Generación de Conocimiento (Knowledge Generation Projects), Spain
Role: co-applicant, value: 125,380 EUR

Dorothy Bishop Award, UK Reproducibility Network (2025)
Early Career Researcher award for promoting and improving open research practices

Elucidating diversity dynamics in marine tropical hotspots (EDiTH) (2024)
Natural Environment Research Council Independent Research Fellowship, UK
Role: applicant, value: 874,978 GBP

Addressing the origins of marine invertebrate diversity in computer simulations (INDICiOS) (2024)
Proyectos de Generación de Conocimiento (Knowledge Generation Projects), Spain
Role: co-applicant, value: 345,000 EUR

Reconstructing the diversity dynamics and drivers of Cenozoic marine biodiversity hotspots (2024)
Norman Newell grant from the Paleontological Society, USA
Role: applicant, value: 5000 USD

rmacrostrat R package development workshop (2024)
Scientific Exchange grant from the Swiss National Science Fund, Switzerland
Role: co-applicant, value: 5800 CHF

MURKY: Mesophotic and turbid reefs as key ecosystems for the future? (2023)
Research workshop grant from the Paleosynthesis project, Friedrich-Alexander-Universität, Germany
Role: co-applicant, value: 10,000 EUR

R for Palaeobiologists: Workshop and Hackathon (2023)
Meeting support grant from The Paleontological Society, USA
Role: co-applicant, value: 1000 USD

R for Palaeobiologists: Workshop and Hackathon (2022)
Grant-in-Aid from The Palaeontological Association, UK
Role: applicant, value: 2000 GBP

Juan de la Cierva Postdoctoral Research Fellowship (2022)
Agencia Estatal de Investigación, Spain
Role: applicant, value: 64,800 EUR

Doctoral Summer School for Advanced Spatial Modelling - Hackathon award (2019)
Awarded for best hackathon project, Centre for Advanced Spatial Analysis, University College London, London
Value: 250 GBP

13th International Symposium on Fossil Cnidaria and Porifera - Travel grant (2019)
Travel grant from The Palaeontological Association, UK
Role: applicant, value: 220 EUR

President's PhD Scholarship (2016)
Imperial College London, London, UK
Role: applicant, value: 79,000 GBP

Roger Ward thesis award (2015)
Awarded for best Master's thesis, University of Hull, Hull, UK

Value: 50 GBP

Invited presentations

Department of Earth Sciences, University College London, London, UK (2025)
Talk: Open Science, Software, and Coding

Centre for Biodiversity and Environment Research, University College London, London, UK (2024)
Talk: Mind the Gap: Reconstructing marine biodiversity through time and space

Computational Palaeobiology Mini-Symposium (2024)
Department of Biosystems Science and Engineering, ETH Zürich, Basel, Switzerland
Talk: Mind the uncertainty: Global Plate Model choice impacts deep-time palaeobiological studies

Methods in Ecology and Evolution: Live! (2023)
British Ecological Society, Online workshop
Talk: Introduction to rphylopic: An R package for fetching, transforming, and visualising PhyloPic silhouettes

Department of Palaeontology, University of Vienna, Austria (2023)
Talk: Into the palaeoverse: a community-driven R package

PalaeoSIG (British Ecological Society): Masterclass in R (2022)
Talk: Into the palaeoverse: a community-driven R package

Getech PLC (2018)
Talk: Ecological modelling in Geosciences

Selected conference presentations

4th Crossing the Palaeontological-Ecological Gap Meeting (Zürich, Switzerland, 2025)
Poster: Elucidating Diversity Dynamics in Marine Tropical Hotspots: Caribbean

68th Annual Meeting of the Palaeontological Association (Erlangen, Germany, 2024)
Poster: rmacrostrat: An R package for retrieving data from the Macrostrat geological database

12th North American Paleontological Convention (Ann Arbor, Michigan, USA, 2024)
Talk: Mind the uncertainty: Global Plate Model choice impacts deep-time palaeobiological studies
Poster: palaeoverse: A community-driven R package to support palaeobiological analyses
Poster: rmacrostrat: An R package for retrieving data from the Macrostrat geological database

14th International Fossil Coral and Reef Society symposium (Chęciny, Poland, 2023)
Talk: Inverse latitudinal diversity gradients in Last Interglacial coral reef communities

The Palaeontological Association: Annual Meeting (Cork, Ireland, 2022)
Poster: Niche characterisation is biased by limited and heterogeneous spatial sampling

Simposio de Paleontologia del noroeste peninsular (A Coruña, Spain, 2022)
Poster: palaeoverse: a community-driven R package

The Palaeontological Association: Annual Meeting (Valencia, Spain, 2019)
Talk: Poor spatial sampling coverage obscures our understanding of the latitudinal biodiversity gradient in deep time
Poster: Uneven spatial sampling in Phanerozoic palaeoclimate curves

13th International Symposium on Fossil Cnidaria and Porifera (Modena, Italy, 2019)
Talk: The evolution of the latitudinal biodiversity gradient in zooxanthellate corals

The past is a foreign country: how much can the fossil record actually inform conservation? (London, UK, 2019)
Poster: Looking through the palaeontological window: reducing geographical and environmental bias in forecasting biodiversity responses

Crossing the Palaeontological–Ecological Gap (Leeds, UK, 2018)
Talk: Absence of evidence is not evidence of absence: quantifying reef coral range shifts under global climatic change

British Ecological Society - Macroecology (St. Andrews, UK, 2018)
Talk: Anthropogenic climate change and transitional distribution ranges of reef corals: insights from the last Pleistocene interglacial

The Palaeontological Association: Progressive Palaeontology (Manchester, UK, 2018)
Talk: Coupling palaeontological and neontological data: transitional distribution ranges of reef corals under global climatic change

The Palaeontological Association: Annual Meeting (London, UK, 2017)
Poster: Ecological niche modelling in deep time: constraining the transitional distribution ranges of reef corals

European Coral Reef Symposium (Oxford, UK, 2017)
Poster: Anthropogenic climate change and transitional distribution ranges of reef corals: insights from the last Pleistocene interglacial

The Palaeontological Association: Progressive Palaeontology (Leicester, UK, 2017)
Poster: PalaeoENM: Ecological Controls on Triassic/Jurassic Coral Reefs

Fieldwork experience

El Qoseir, Egypt (2023)
Modern and fossil (Last Interglacial) coral reef data collection via line-intercept transects

León, Spain (2022)
Logging and bulk sampling of Devonian sequences

Colorado and New Mexico, USA (2016)
Logging, sampling, and fossil prospecting across the Cenomanian/Turonian boundary

Somerset, Lyme Regis, and Newark, UK (2015)
Logging, sampling, and geochemical analysis across the Triassic/Jurassic boundary

Koh Rong, Cambodia (2012)
Marine biological surveys as part of Koh Rong's (Cambodia) coral reef monitoring project

Professional development

Fair and Inclusive Recruitment (2024)
University College London, London, UK

Introduction to Research Supervision (2024)
University College London, London, UK

Macroevolution and Phylogenies (2022)
Transmitting Science, Online

Introduction to GEN3SIS (2022)
Universidade de Vigo, Vigo, Spain

Towards more equitable marine research (2021)
Centro de Investigación Mariña (led by Transmitting Science), Universidade de Vigo, Spain

Data Manipulation and Statistical Analysis in Palaeoecology (2021)
Palaeoecology Group, British Ecological Society, Online

Doctoral Summer School for Advanced Spatial Modelling (2019)
The Centre for Advanced Spatial Analysis, University College London, London, UK

Species distribution modelling: fundamentals and the future workshop (2019)
University of Nottingham, Nottingham, UK

Advanced ecological niche modelling (2018)
University of Porto, Porto, Portugal

Introduction to ecological niche modelling (2017)
Transmitting Science, Barcelona, Spain

Introduction to coral identification (2017)
European Coral Reef Symposium, Oxford, UK

Introduction to larval dispersal modelling (2017)
European Coral Reef Symposium, Oxford, UK

Student supervision

Jason Wang (2025–Current)
University College London, UK
PhD project: The taxonomic and ecomorphological diversity of birds through deep time

Gloria Herrero (2025–Current)
Institut de Ciències del Mar, Spain
PhD project: Addressing the origins of marine invertebrate diversity in computer simulations

Xiaoli Ma (2024–2025)
Nanjing University, China
PhD project: Palaeogeographic reconstructions shape understanding of deep-time climate change

Die (Wendy) Wen (2024–2025)
University College London, UK
PhD project: Reconstructing Palaeozoic marine latitudinal diversity gradients

Theres Koch (2024)
University of Vienna, Austria
Master's project: Checklist of scleractinian species in Mangrove Bay, El Qoseir, Egypt

Miranta Kouvari (2019–2023)
University College London, London, UK
Master's project: The latitudinal biodiversity gradient of South American eutherians

Lucas Buffan (2021–2022)
University of Vigo, Vigo, Spain
Master's project: Spatial discrepancies in palaeogeographic reconstructions

Teaching and pastoral care

Guest Lecturer (2024–Current)
Department of Earth Sciences, University College London, UK

- The Earth - Meet Your Researcher (GEOL0007)
- Vertebrate Palaeontology and Evolution (GEOL0009)
- Advanced Biodiversity and Macroevolutionary Studies (GEOL0036)

Building Open Data Science Skills in Palaeobiology and Ecology (2025)
4th Crossing the Palaeontological-Ecological Gap Meeting, Zürich, Switzerland
Organised and ran a workshop on open data science skills

R for Paleobiologists: Getting Started with the palaeoverse R Package (2024)
12th North American Paleontological Convention, Ann Arbor, Michigan, USA
Organised and ran a workshop on using R in Palaeobiology

Lab Assistant (2024)
Universidade de Vigo, Vigo, Spain

- Computational Tools in Biology

R for Palaeobiologists: Workshop and Hackathon (2023)
University College London, UK
Organised and ran a workshop on using R in Palaeobiology

Graduate Teaching Assistant (2016–2021)
Imperial College London, London, UK

- Palaeobiology I
- Palaeobiology II
- Dorset (UK) geology field course
- Almería (Spain) geology field course

Student Hall's Pastoral Warden (2016–2021)
Woodward Buildings, Imperial College London, London, UK

Professional appointments, service, and outreach

IFCRS 4th Early Career Researcher Symposium (2025)
International Fossil Coral and Reef Society, Online
Co-organised and ran an early career researcher meeting on fossil corals and reefs

Early Career Researcher Forum, Faculty of Mathematical & Physical Sciences (2025–Current)
Early Career Researcher representative for the Department of Earth Sciences

Generation Hope, Natural History Museum, London, UK (2025)
Outreach science stall on coral resilience

Scientific Associate, Natural History Museum, London, UK (2025–Current)
Palaeontology, Department of Earth Sciences

Handling Editor for *Open Palaeontology* (2024–Current)
Open Palaeontology is a new international Diamond Open Access journal promoting open science in the field of palaeontology

68th Annual Meeting of the Palaeontological Association (Erlangen, Germany, 2024)
Session chair: Climate Change

Recent Advances in Computational Paleobiology symposium (Ann Arbor, Michigan, USA, 2024)
Co-organised and ran a symposium at the 12th North American Paleontological Convention

Palaeoverse Lecture Series (2024–Current)
Monthly lecture series on recent advances in computational tools and methods in palaeobiology **https://www.youtube.com/@ThePalaeoverse/**

14th International Fossil Coral and Reef Society symposium (Chęciny, Poland, 2023)
Session chair: Research Methods

IFCRS Internet Officer (2023–current)
Council member of the International Fossil Coral and Reef Society (IFCRS)

IFCRS Early Career Researcher Representative (2021–2023)
Council member of the International Fossil Coral and Reef Society (IFCRS)

IFCRS Lecture Series (2022)
International Fossil Coral and Reef Society, Online
Co-organised and ran a lecture series on fossil corals and reefs

IFCRS 1st Early Career Researcher Symposium (2021)
International Fossil Coral and Reef Society, Online
Co-organised and ran an early career researcher meeting on fossil corals and reefs

Fossil Fact or Fiction stall (2019)
Einstein's Garden, Green Man Festival, Brecon Beacons, Wales
Co-ran a stall on fossil record bias over the course of the festival

Ecological niche modelling workshop (2018)
Imperial College London, London, UK
Co-organised and ran a workshop on ecological niche modelling

Palaeontological Association meeting (2017)
Imperial College London, London, UK
Co-organised and ran the annual meeting of the Palaeontological Association

Peer-review experience

Publications
Big Earth Data Journal (1), *Conservation Biology* (1), *Ecological Monographs* (1), *Global Ecology and Biogeography* (3), *Journal of Biogeography* (1), *Palaeontology* (2), *Paleobiology* (3), *Methods in Ecology and Evolution* (4), *Nature Communications* (1), *Science Advances* (1), *Scientific Reports* (1), *Trends in Ecology and Evolution* (1)
Research proposals
Daphne Jackson Fellowship (1)

PhD Theses
External examiner: Charlotte Clay, University of Leeds (2025)

Skills

R programming
Advanced proficiency in data analysis, data visualization, writing functions and developing packages

Geocomputation
Proficient in spatial analysis and the use of R spatial packages (e.g. raster, terra, sf, sp)

Languages
Spanish (CEFR B1)
Mandarin Chinese (HSK1)

SCUBA diver
BSAC-accredited Ocean and Sports Diver

Workplace qualifications
Emergency first-aid at work & automated external defibrillator
Outdoors emergency first-aid
Mental health first-aid

Fire safety and warden
Active-bystander

Driving
Full EU driving license

Memberships

Member of the British Ecological Society (2017–present)
Member of the Palaeontological Association (2017–present)
Member of the International Fossil Coral and Reef Society (2018–present)
Member of the Conservation Paleobiology Network (2019–present)
Member of the Paleontological Society (2023–present)

Research statement

Current research plans

I am currently a NERC Independent Research Fellow (2024–2029) working on the project 'Elucidating Diversity Dynamics in Marine Tropical Hotspots' (EDiTH). Today, global marine biodiversity peaks in the Indo-Australian Archipelago, yet the fossil record suggests that the locations of marine biodiversity hotspots have shifted repeatedly, with at least four distinct centres identified during the Cenozoic (last 66 million years). For my fellowship, I am reconstructing the evolutionary history of these biodiversity hotspots and testing the mechanisms that drove their emergence, persistence, and decline. To do so, I am combining Earth System Modelling with palaeontological data (corals) to perform state-of-the-art macroecological and macroevolutionary modelling.

In addition to my current research focus, I am co-leading a large interdisciplinary working group with Dr Nadia Santodomingo (Senckenberg Research Institute and Natural History Museum, Germany) through the 'Mesophotic and turbid reefs as key ecosystems for the future?' (MURKY) project. Today, increasingly frequent and intense marine heatwaves are driving widespread coral mortality within the euphotic zone. In contrast, mesophotic coral ecosystems often show greater resilience, leading to the hypothesis that they may serve as refugia under accelerating climate change. Yet, whether mesophotic environments offer long-term refuge remains unclear. Through the MURKY project, I am aiming to address this research gap by bridging neontological and palaeontological perspectives to understand whether mesophotic environments served as refugia for coral communities during past climatic change events (e.g. hyperthermals).

Through partnership with Dr Kenneth Johnson (Natural History Museum, UK) and George Oates (Flickr Foundation), I am currently co-leading the 'Archival Reef Imagery for Ecological Legacies' (ARIEL) project. This project is focused on building a global database of historical photographs of coral reefs to document their decline, and to help address the socio-psychological phenomenon of 'shifting baselines'. The project is currently in its first phase, focused on public engagement through imagery. After this initial phase, we intend to use these historical archives in combination with AI-driven image recognition software to extract ecologically-relevant data (e.g. presence/absence of species, changes in fish sizes/populations) and quantify temporal shifts.

Finally, I am the founder of Palaeoverse (**https://palaeoverse.org**), an organisation aiming to bring the palaeobiological community together to share resources, reach agreed standards, and improve reproducibility in the discipline. To date, this has led to the generation of various community resources including software toolkits such as the palaeoverse R package. Recently, I secured a large award from the Software Sustainability Institute's Research Software Maintenance Fund to carry out the project 'Converting Users to Contributors: Enabling Sustainable Maintenance and Development of Palaeoverse' (2026–2028). The project will strengthen the Palaeoverse ecosystem by improving package maintainability, expanding contributor documentation, delivering community training and mentorship for early-career researchers, and reforming governance to support long-term, community-led sustainability.

Future research plans

Over the next five years, I plan to build and lead my own research group focused on deep-time marine macroecology and macroevolution, expanding the scope of my research in two directions. First, I plan to submit a European Research Council Starting Grant proposal next year focused on modelling the dispersal of marine organisms across spatial (local, regional, global) and temporal scales (past, present, and future), including geological timescales. The project will develop novel tools, frameworks, and resources for simulating dispersal, and will assess how key drivers–such as climate change, sea-level fluctuations, and tectonic processes–influence dispersal dynamics and,

ultimately, shape biodiversity patterns (e.g. the emergence of biodiversity hotspots). This research will further strengthen my active collaborations with Earth System modellers within the School of Geographical Sciences (e.g. Prof. Paul Valdes, Prof. Dan Lunt, Dr Alexander Farnsworth) at the University of Bristol. Looking further ahead (5+ years), I aim to pursue an ERC Synergy Grant to establish a large, international research network to evaluate the generality of biodiversity trajectories (past, present, future) within the Indo-Australian Archipelago marine biodiversity hotspot across multiple taxonomic groups. I will also continue to actively support prospective PhD students and postdoctoral research fellows. For example, I am currently co-developing projects to work on themes such as the evolution of photosymbiosis in corals and leveraging AI-driven data mining tools for historical ecology and conservation palaeobiology. Over the next ten years, I aim to cement myself as a global research leader in marine macroecology and macroevolution. At that stage, I envisage pursuing significant funding opportunities such as Leverhulme Research Centres or Leverhulme Doctoral Scholarship to further reshape the field and transform our understanding of marine macroecology and macroevolution. In summary, I aim to dedicate my research to investigating how biodiversity responds to environmental change across timescales, from the deep past to the Anthropocene.

Teaching statement

My teaching philosophy is rooted in fostering active learning, student motivation, and a supportive environment. I believe that learning is most effective when it is student-centred, grounded in constructivist principles that view knowledge as something students actively build through meaningful engagement. By embedding active learning across all teaching contexts–lectures, laboratories, tutorials, and fieldwork–I aim to inspire curiosity, critical thinking, and a lifelong passion for learning. I strive to create a flexible and inclusive learning environment that recognises the diverse backgrounds, skills, and motivations of students. This includes tailoring discussions to highlight the broader relevance of topics, encouraging intrinsic motivation, and clearly communicating learning outcomes and transferable skills. **My goal is to equip students with the ability to teach themselves and others, fostering independence, confidence, and collaboration**.

I have extensive teaching experience across undergraduate and postgraduate levels in Life and Earth Sciences at four universities, in three different countries. My roles have encompassed leading small-group tutorials, supervising laboratory practicals, demonstrating in the field, running seminars and workshops, delivering guest lectures, and engaging wider audiences through public outreach. My teaching primarily focuses on palaeobiology, ecology, programming, and open science skills. This diverse experience reflects my capacity and enthusiasm for engaging in a broad range of teaching contexts.

During my PhD at Imperial College London (2016–2020), I served as a Graduate Teaching Assistant in the Department of Earth Science and Engineering, where I supported palaeobiology courses and geology field trips, such as those in Almería, Spain. As a postdoctoral researcher (2021–2024), I contributed to teaching the 'Computational Tools in Biology' course at the University of Vigo (Spain) and the 'Conservation Palaeobiology and Historical Ecology' field course in El Qoseir, Egypt, for the University of Vienna (Austria). In my current fellowship (2024–Present), I design and deliver original content and teaching materials for guest lectures in modules such as 'Vertebrate Palaeontology and Evolution' and 'Advanced Biodiversity and Macroevolutionary Studies' at University College London. Through these varied experiences, I have demonstrated my ability to design, deliver, and support teaching across different modules and settings.

In assessing student progress, I employ a range of formal and informal techniques. These include marking reports, using reflective exercises such as identifying the 'muddiest point' from lecture materials, and posing targeted questions to gauge understanding and guide further learning. I prioritise providing constructive, targeted feedback to support continuous improvement. In small-group tutorials, this involves group discussions of challenging topics, encouraging students to explain concepts to one another, and fostering peer-to-peer learning. I find this approach helps students monitor their learning, promoting self-directed and independent study.

Postgraduate supervision has been a large aspect of my teaching experience to date. I have co-supervised three Master's students and currently co-supervise four PhD student projects, providing guidance on research design, data analysis, scientific writing, and career planning. I place equal emphasis on technical training and professional growth, encouraging students to publish their research, present at conferences, and develop transferable skills. This approach has yielded strong results to date, with two Master's students publishing their research in *Methods in Ecology and Evolution* and *Biodiversity Data Journal*, while a third has a manuscript currently under review in *Current Biology*. I would highly value the opportunity to supervise further at the University of Bristol, and actively support the development of individuals on a one-to-one basis.

Beyond formal university teaching, I actively engage in training the wider research community, particularly early career researchers. Over the past three years, I have developed and led multiple workshops on programming and open science skills at major palaeontological and ecological conferences, including most recently at the 'Crossing the Palaeontological–Ecological Gap' meeting in Zurich, Switzerland (2025). Additionally, I have delivered invited online lectures and workshops for organisations such as the British Ecological Society's 'MEE Live!' series and PalaeoSIG's 'Masterclass in R'. Outside academic settings, I have also engaged in a number of outreach activities,

including running public science stalls (e.g. Generation Hope, Natural History Museum; Einstein's Garden, Greenman Music Festival). If appointed to this position, I would take great pride in contributing to the School of Biological Sciences' outreach and engagement activities, especially those involving local schools, natural history film makers, and policy makers.

The emphasis on quality undergraduate and postgraduate education at the University of Bristol is a major reason for my interest in this position. I am strongly committed to continuous professional development in teaching and would pursue accreditation from the Higher Education Academy if appointed to this position. I am enthusiastic about contributing to a broad range of existing undergraduate and postgraduate modules within the School of Biological Sciences, such as 'Diversity of Life', 'Evolutionary Biology', 'Ecology and Conservation', 'Quantitative and Computational Methods', 'Practical Research Skills', 'Blue Planet', and 'Understanding Data: Experimental Design and Statistics for Life Scientists', and I am well positioned to contribute to the joint honours MSci in Palaeontology and Evolution. Additionally, I am passionate about developing new modules focused on 'Invertebrate Palaeontology', 'Phanerozoic Reef Evolution', and 'Historical Ecology and Conservation Palaeobiology', with the latter potentially containing a field-based research project. These modules could serve to further bridge the School of Biological Sciences and School of Earth Sciences. Furthermore, I am deeply committed to advancing programming and open science education, which I consider foundational for students' future success in both academia and industry. If appointed, I would relish the opportunity to contribute to existing modules on these themes and to develop new content that supports students in building robust, transferable skills. Through these contributions, I would aim to inspire curiosity, critical thinking, and a lifelong passion for learning among my students, while supporting the University of Bristol's commitment to excellence in education.